Diseases of Amaranthus, spinach, Sorrel and Portulaca

1. Damping off (amaranthus, spinach) - *Pythium* spp
2. White rust (Amaranthus) - *Albugo bliti*
   Spinach - *Albugo occidentalis*
   Portulaca - *A. portulaceae*
3. Downy mildew (spinach) - *Pernospora spinaciae*
4. Anthracnose (spinach) - *Colletotrichum spinacicola*, *Colletotrichum dematium* f.sp *spinaciae*
5. Cercospora leaf spot (amaranthus) - *Cercospora brachiata*
6. Phyllosticta leaf spot (amaranthus) - *Phyllosticta chenopodii*
7. Wilt (spinach) - *Fusarium oxysporum* f.sp.*spinaciae*
8. Rust (Spinach) - *Puccinia aristidae*
   (Sorrel) - *P. oxyriae*
9. Mosaic (Spinach) - Cucumber Mosaic Virus

Damping off - *Pythium aphanidermatum, Rhizoctonia solani, Aphanomyces* sp.

Damping-off, *Pythium aphanidermatum*, *Rhizoctonia solani*, *Aphanomyces* sp.

Damping-off, *Pythium aphanidermatum*, *Rhizoctonia solani*, *Aphanomyces* sp.

Symptoms:

- Damping-off is a serious disease in seedlings.
- The disease is caused by *Pythium aphanidermatum*, *Rhizoctonia solani* and *Aphanomyces* sp.
- Seeds may rot in the soil before emergence (pre-emergence damping-off) or seedlings may exhibit **stem canker above the soil line and/or root necrosis.**

Damping-off, *Pythium aphanidermatum*, *Rhizoctonia solani*, *Aphanomyces* sp.

Symptoms contd…

- Affected seedlings eventually wilt (post-emergence damping-off).
- The disease is favoured by **high soil water content and low soil temperatures.**
- Also **dense planting without sufficient aeration** enhances disease development.

Damping-off, *Pythium aphanidermatum*, *Rhizoctonia solani*, *Aphanomyces* sp.

Management

- Use disease free seeds
- Avoid over watering
- Avoid dense sowing and planting
- Provide good drainage
- Drenching with COC

Amaranthus White rust- *A.bliti*

Portulaca - *A. portulaceae*

White rust of spinach- *Albugo occidentalis*

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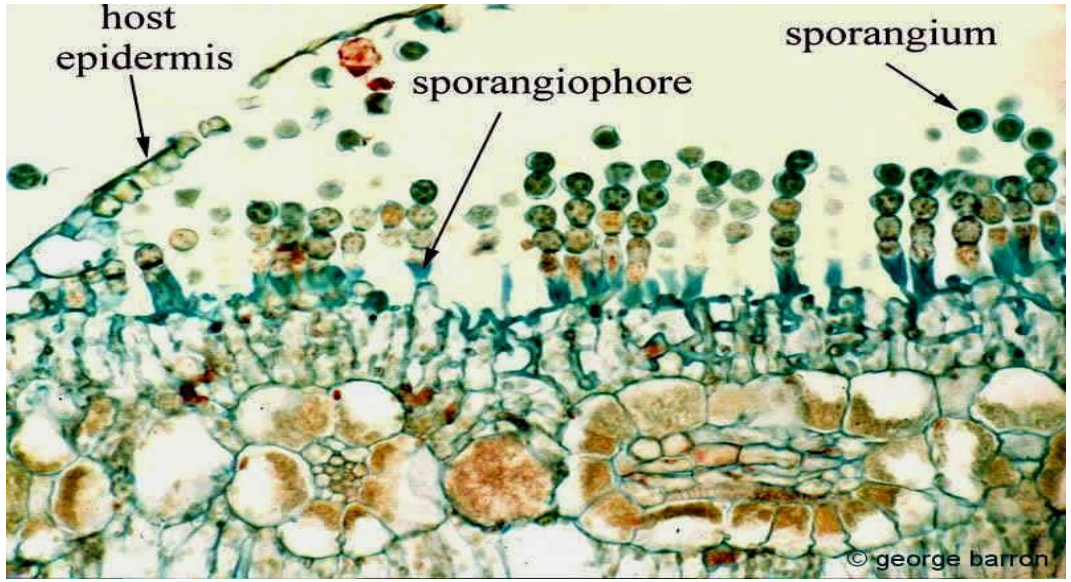
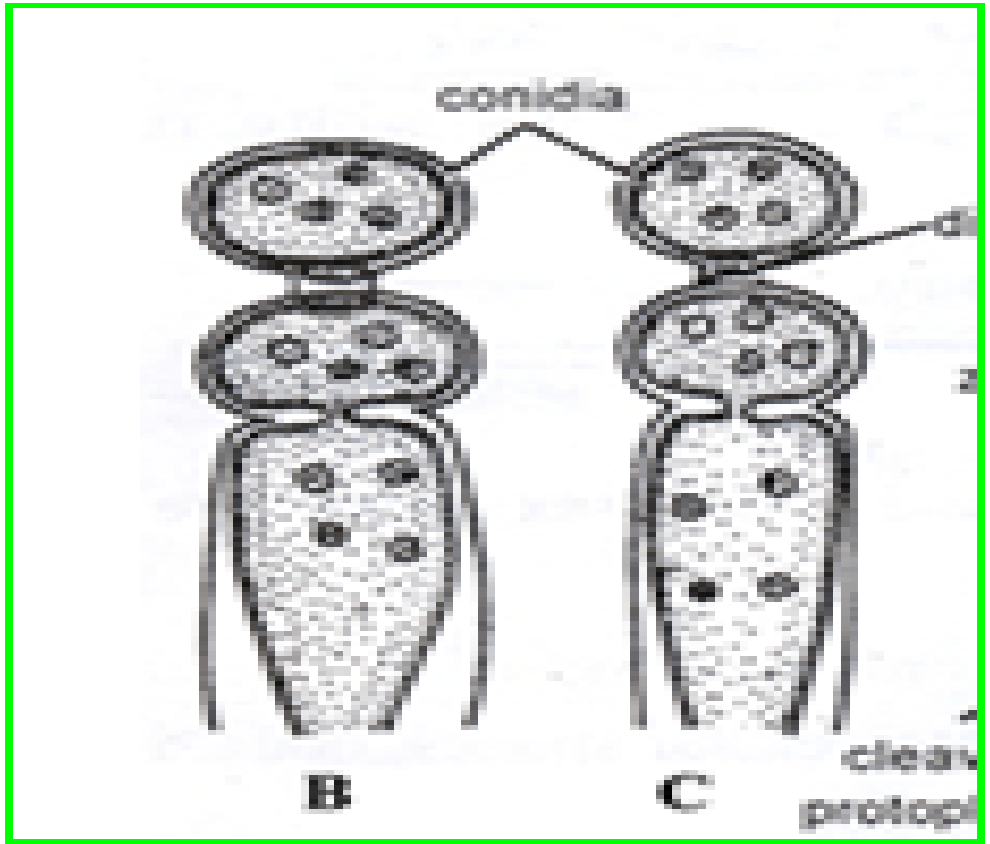
White rust of spinach- *Albugo occidentalis*

Symptoms

- Raised white pustules form on the undersides of infected leaves and often coalesce.
- The appearance of the upper leaf surface is typically light yellow.
- **Anthocyanin may be produced around the sori** resulting in reddish violet blotches that are most prominent on the upper leaf surface.
- The disease drastically reduces quality and marketability of the leaves.

- The sporangiophores are short, hyaline, unbranched, thick walled, basally branched, determinate growth club-shaped and give rise to simple chains of sporangia.
- Sporangiophores arranged in palisade manner
- Sporangia formed in **basipetal succession.**
- The accumulating sporangia rupture the epidermis and white or creamy white mass appears (White rust or blisters or pustules)





Spinach Downy mildew - *Perenospora farinosa* f.sp. *spinaciae, P. effusa*

Spinach Downy mildew - *Perenospora farinosa* f.sp. *spinaciae, P. effusa*

Downy mildew -*Peronospora farinosa* f.sp. *spinaciae*

Symptoms

- Disease is most important at the seedling stage.
- The cotyledons are systemically infected, becoming discoloured and distorted.
- Loss of seedlings causes uneven crop development.

Disease cycle

- The pathogen persists as oospores in the soil, or on beet seed crops, or on overwintered volunteer beet plants

Cercospora leaf spot - *Cercospora canescens*

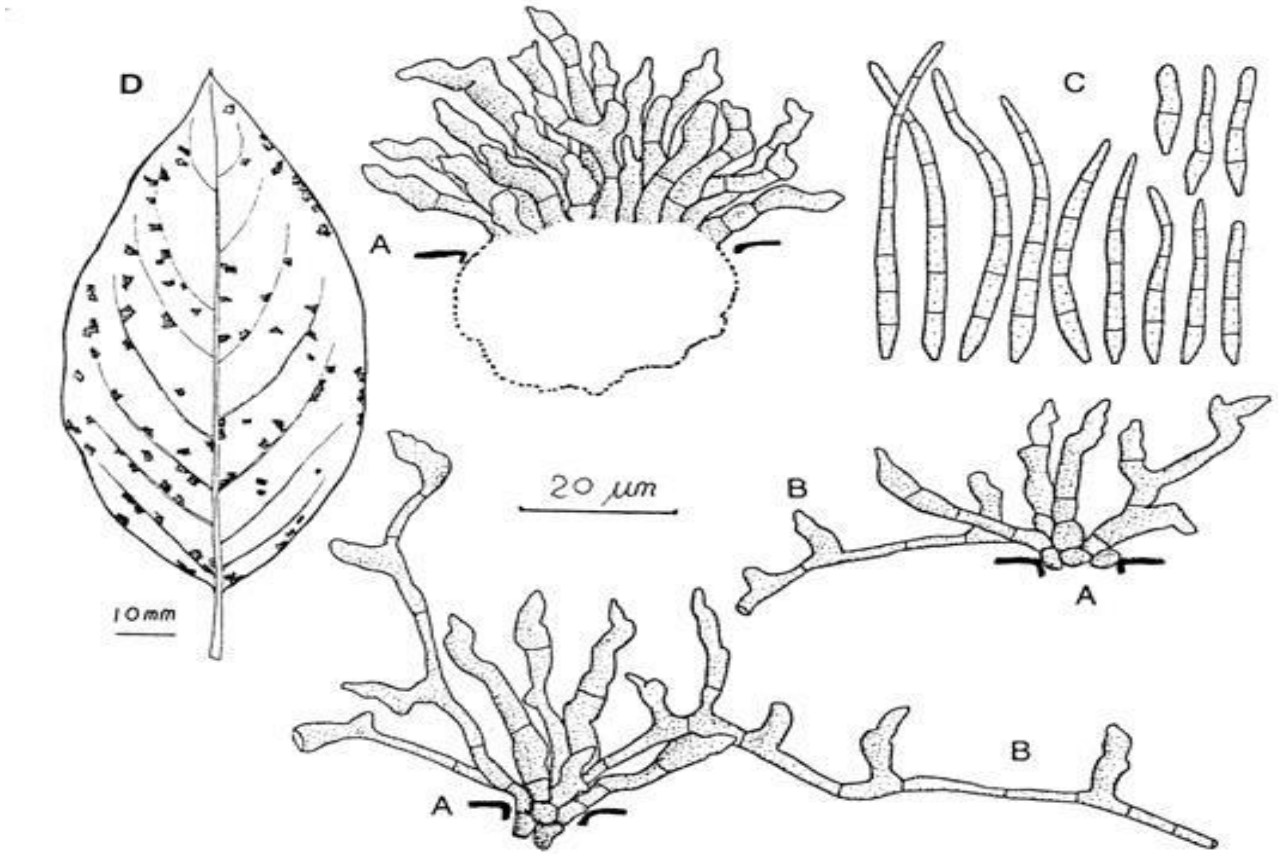
Symptoms

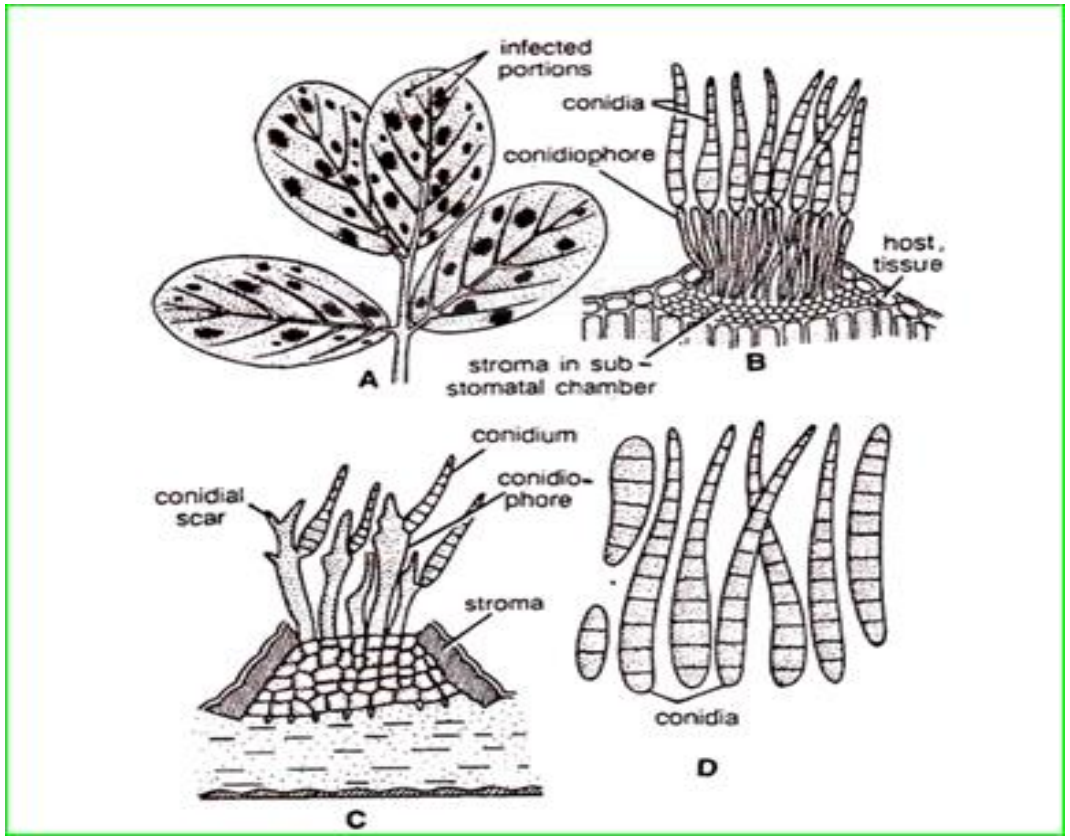
- **Rough, circular, white coloured** on the upper leaf surface
- The centre becomes **silvery grey when spores** are produced.
- In severe cases, **defoliation occurs**.

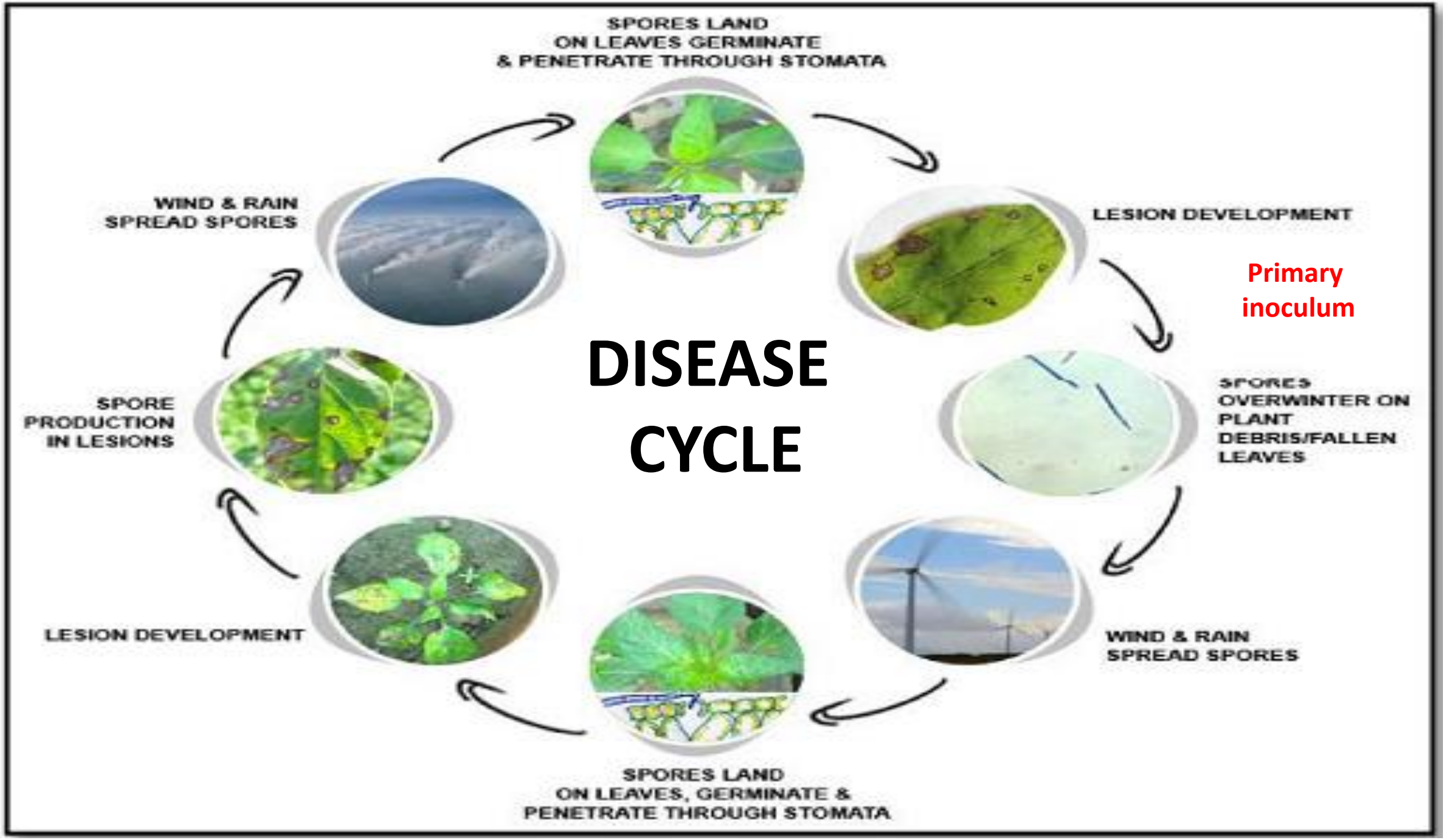
Management:

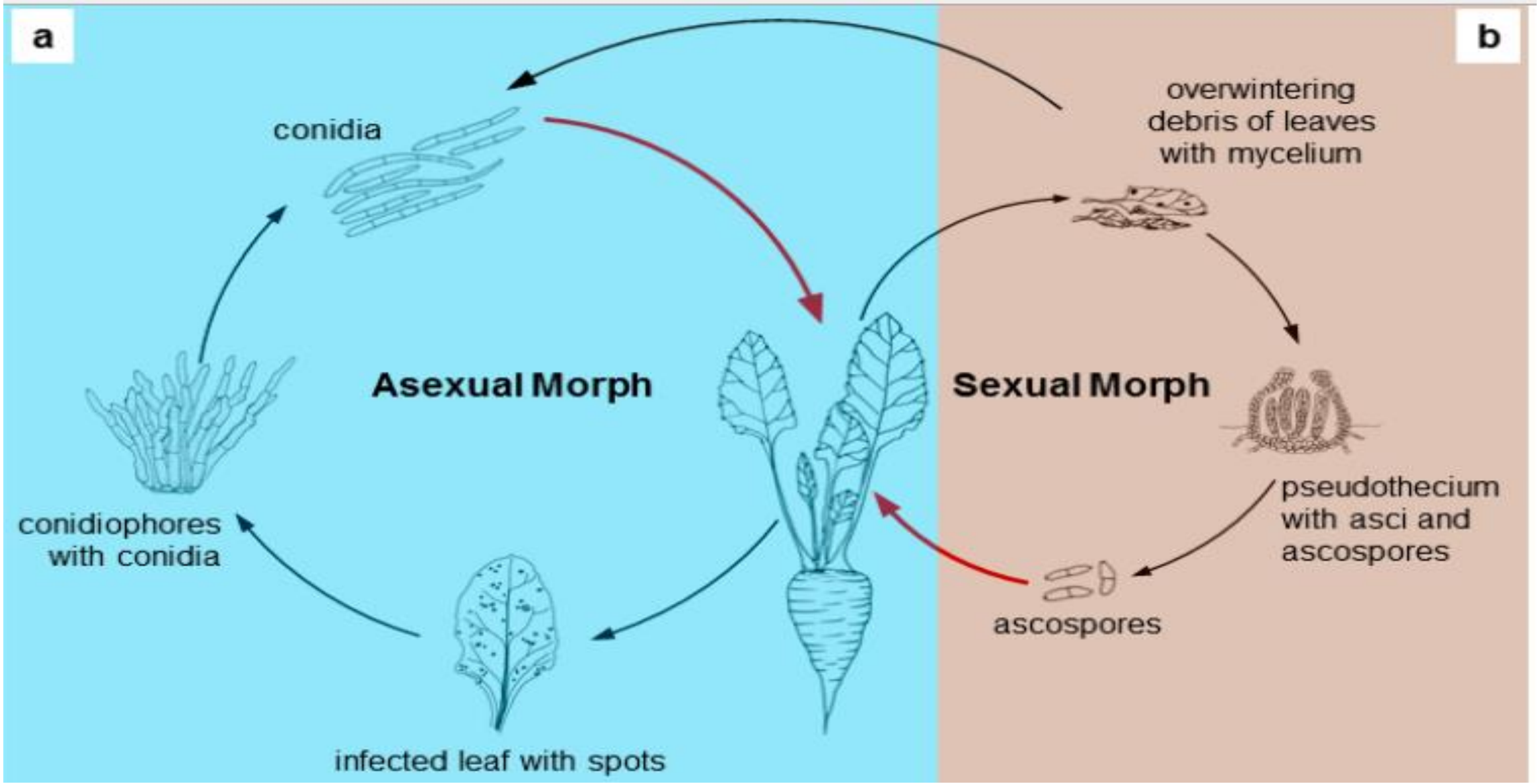
- Destroy diseased debris.
- Use **pathogen free seeds** from healthy plants.
- Seed treatment with carbendazim or thiram @2g/kg of seeds.

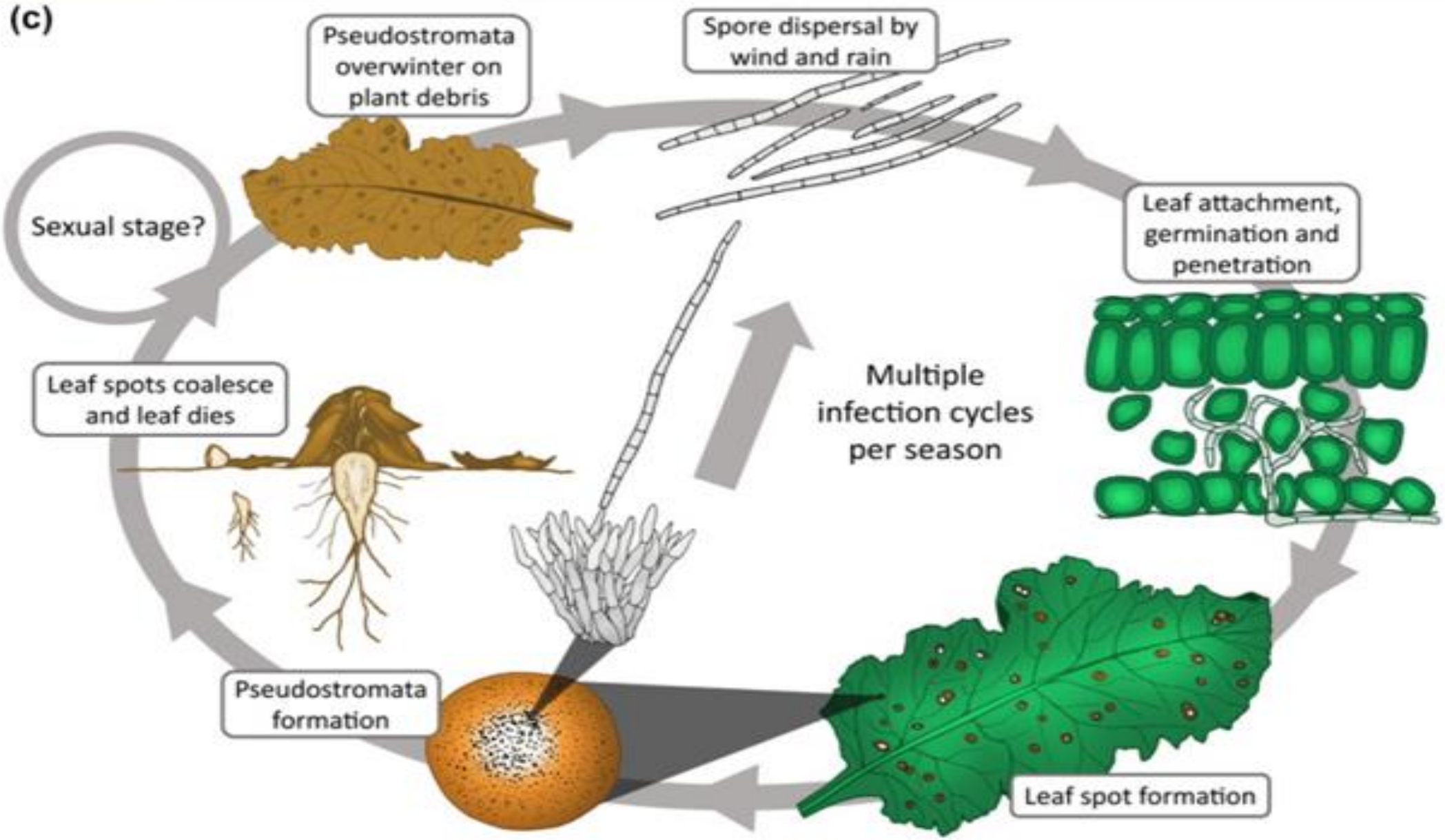
Aetiology or Etiology






SPORES LAND
ON LEAVES GERMINATE
& PENETRATE THROUGH STOMATA
LESION DEVELOPMENT
Primary inoculum
SPORES OVERWINTER ON PLANT DEBRIS/FALLEN LEAVES
WIND & RAIN SPREAD SPORES
SPORES LAND
ON LEAVES, GERMINATE &
PENETRATE THROUGH STOMATA
LESION DEVELOPMENT
SPORE PRODUCTION IN LESIONS
WIND & RAIN SPREAD SPORES
DISEASE CYCLE


a
b
conidia
overwintering
debris of leaves
with mycelium
Asexual Morph
Sexual Morph
conidiophores
with conidia
pseudothecium
with asci and
ascospores
ascospores
infected leaf with spots


(c)
Pseudostromata overwinter on plant debris
Spore dispersal by wind and rain
Leaf attachment, germination and penetration
Sexual stage?
Multiple infection cycles per season
Leaf spots coalesce and leaf dies
Pseudostromata formation
Leaf spot formation

Spinach Anthracnose –*Colletotrichum spinacicola, Colletotrichum dematium* f.sp *spinaciae*

Water soaked lesions on spinach leaves caused by ***Colletotrichum dematium*.**

Spinach Anthracnose –*Colletotrichum spinacicola, Colletotrichum dematium* f.sp *spinaciae*

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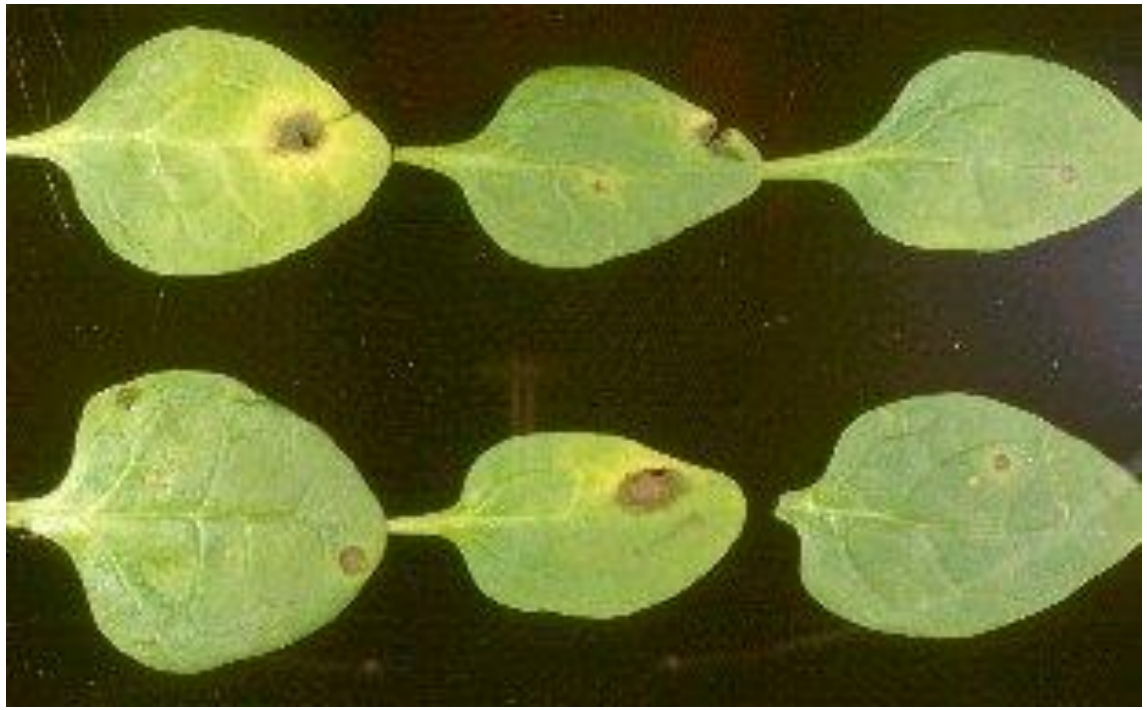
Spinach Anthracnose –*Colletotrichum spinacicola, Colletotrichum dematium* f.sp *spinaciae*



Spinach Anthracnose –*Colletotrichum spinacicola, Colletotrichum dematium* f.sp *spinaciae*

Spinach Anthracnose –*Colletotrichum spinacicola, Colletotrichum dematium* f.sp *spinaciae*

Symptoms :

- Brown sunken lesions with dark red margins develop on the stem.
- **Many lesions rapidly merge to girdle stems, branches, peduncles and petiole**.
- **Elongated lesions appear on the leaf veins** and tannish red spots with yellow halo on the leaf surface.
- **Lesions similar to those found on the stem** also occur on pods .
- **Severely affected pods are curled** and do not contain normal size seeds.

Spinach Anthracnose –*Colletotrichum spinacicola, Colletotrichum dematium* f.sp *spinaciae*

Management:

- Crop rotation with non-host crops for 2-3 years.
- Seed treatment with carbendazim or thiram @2g/kg of seeds.
- Spraying of mancozeb (0.2%) or carbendazim (0.2%) at fortnightly interval.

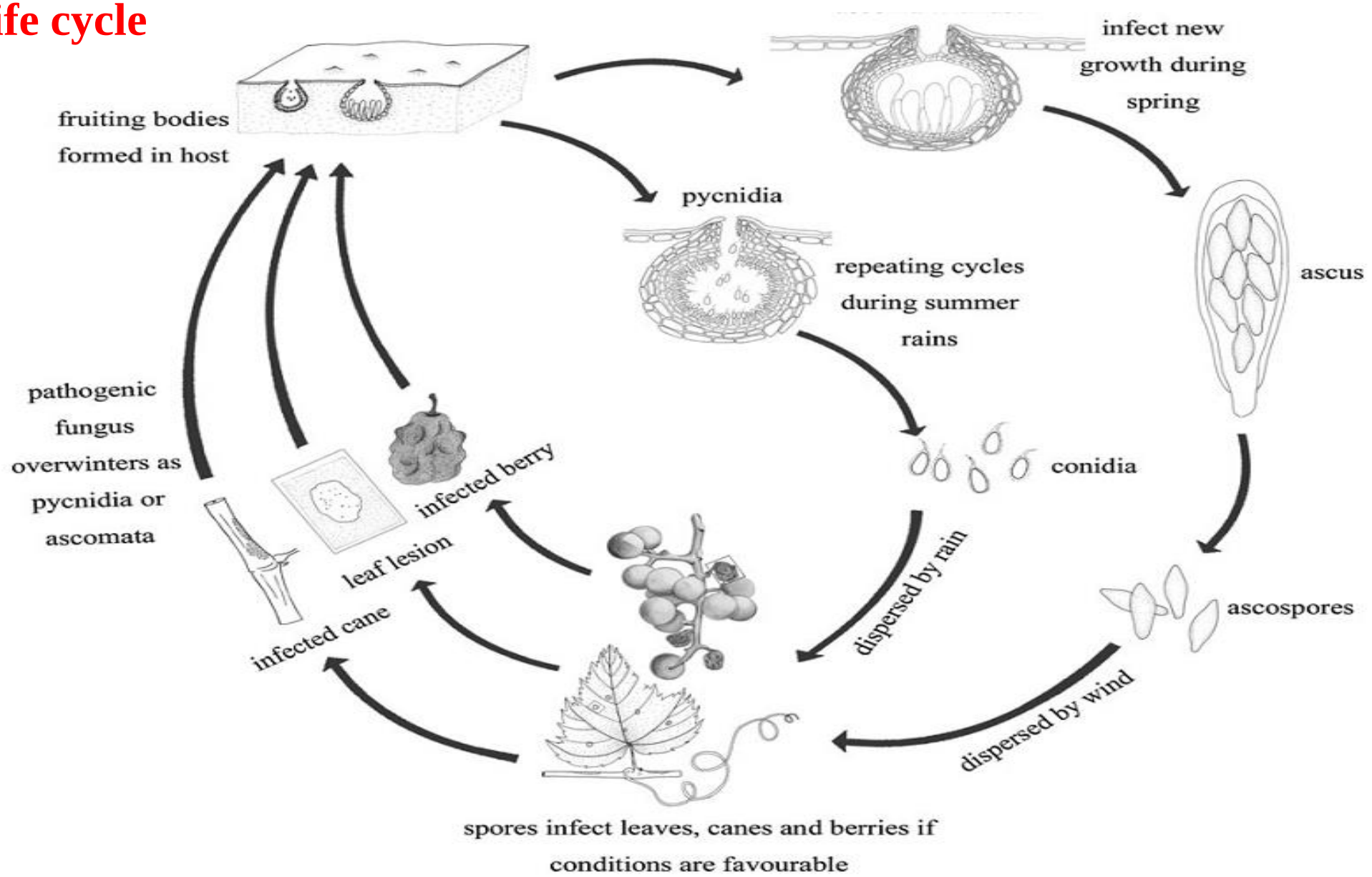
Spinach Phyllasticta leaf spot - *Phyllosticta chenopodii*

Spinach Phyllasticta leaf spot - *P. chenopodii*

Symptoms:

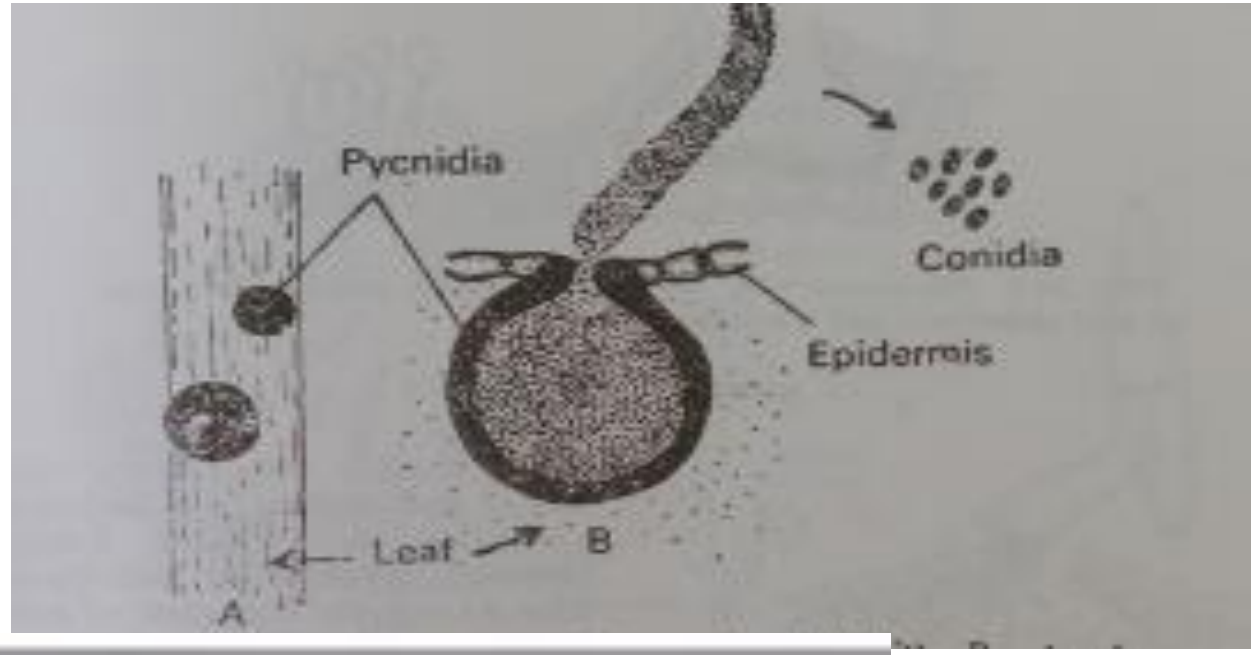
- Circular and brown spots with reddish borders
- Small black pycnidia in the center of the spots
- Dried leaves /susceptible wild plants/on seed

Phyllasticta life cycle



g. 2 Schematic representation of the life cycle of *Phyllosticta* and its teleomorph redrawn from: http://www.oardc.ohio-state.edu/fruitpathol ganic/grape/All-Grapes.html

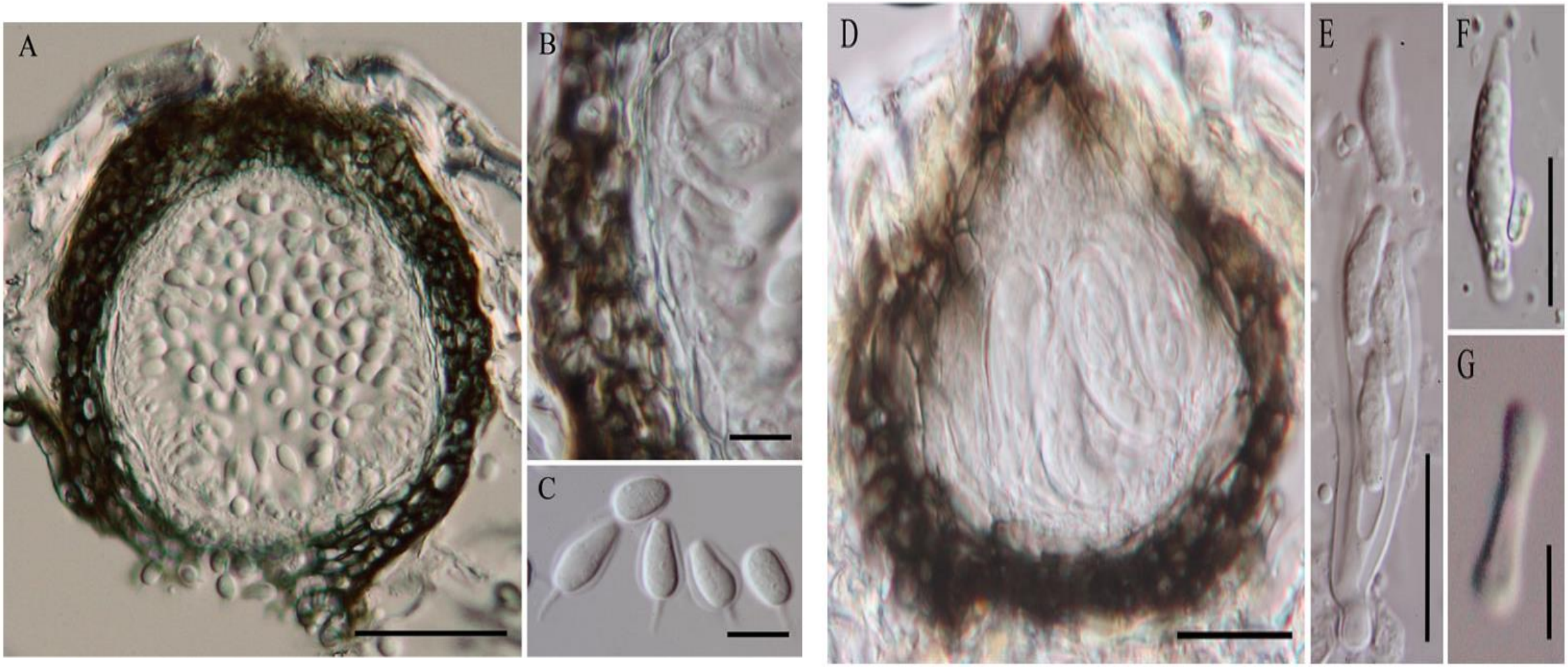
B
Phyllosticta


Pycnidia
Conidia
Epidermis
Leaf
B
A

C

Pycnidia

Ascostroma



Produce simple brown, thin-walled ascomata, containing **bitunicate and fissitunicate asci** and variously shaped, hyaline, unicellular ascospores, with or without mucilaginous sheaths or appendages

Spinach Phyllasticta leaf spot - *P. chenopodii*

Management:

- Use certified seed
- Collect and destroy the crop residue after harvest
- In case of heavy infestation - spray- copper oxychloride (0.25%) harvest the leaves10 days after spraying

Spinach Fusarial wilt- *Fusarium oxysporum* f.sp. *spinaceae*

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Healthy root **Diseased roots**

Spinach Fusarial wilt- *Fusarium oxysporum* f.sp. *spinaceae*

Typical blackening of roots

Spinach Fusarial wilt- *Fusarium oxysporum* f.sp. *spinaceae*

Symptoms:

- The pathogen **affects the crop at any stage of crop growth**.
- **Wilt**: The plants show **yellowing withering and drooping of leaves**.
- The vines show **blackening and drying**.
- The roots show **rotting and blackening**.
- **Basal swelling:** The affected **amaranthus plants became stunted** and weak .

Spinach Fusarial wilt- *Fusarium oxysporum* f.sp. *spinaceae*

Symptoms contd..

- The **base of the stem swells up to the size of a small cassava tuber**.
- The **taproot and lateral roots** are affected.
- The **basal swollen portion becomes disintegrated and shredded**.
- The **stem, leaf petioles become flattened**.
- The **flowers show sterility**.

Spinach Fusarial wilt- *Fusarium oxysporum* f.sp. *spinaceae*

Management

- Avoid planting Fusarium infested soil
- Leafy vegetable 4 to 6 year rotation, seed crop 6- to 12-year rotation
- Early planting
- Smooth-leaf cultivars - **St. Helens and Jade**
- Semisavoy types- **'Chinook II' and 'Skookum'**
- **Amend acid soils (pH < 6.5)** with agricultural limestone @1- 2 t/ac
- **Avoid moisture stress during flowering and seed set.**

Spinach Fusarial wilt- *Fusarium oxysporum* f.sp. *spinaceae*

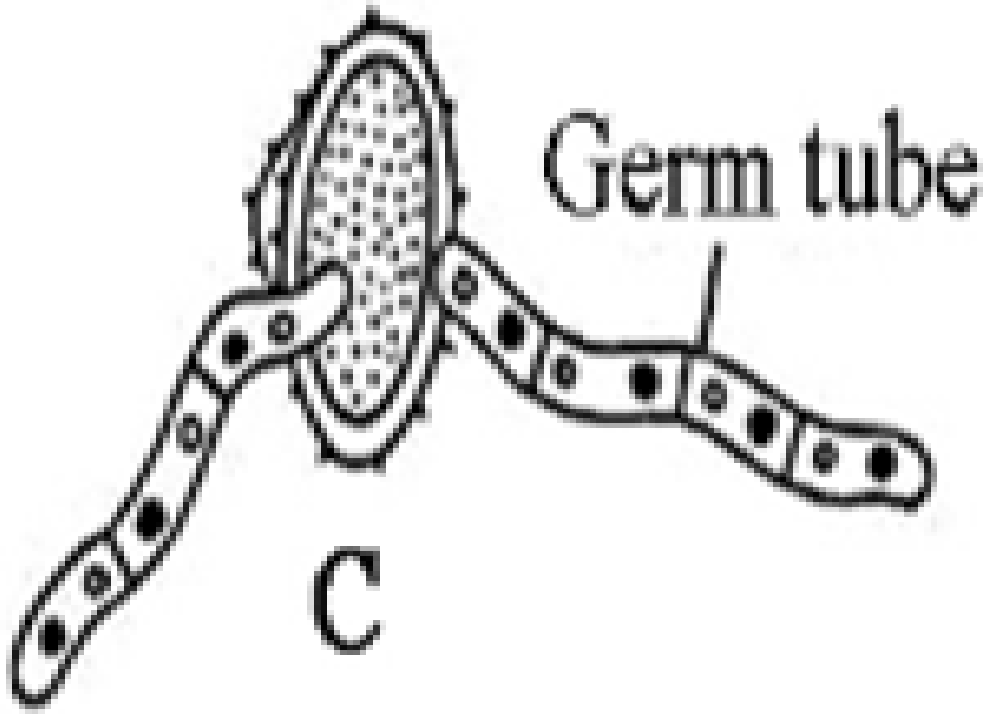
Management contd….

- Ten days before **sowing apply *Trichoderma*** enriched with cow dung and neem cake in the pit.
- Treat the seeds with *Pseudomonas* 1% solution.
- Drench the pit and spray the plant with *Pseudomnas* 2% when the plant attains 15 and 25 days.
- This treatment can be repeated based on need.
- Drenching the base and spraying the plant with copper oxychloride @0.4%.
- There should be a gap of ten days between COC and Pseudomonas application.

Spinach Rust- *Puccinia aristidae*

Young aecia arranged in circular patterns on the underside

Spinach Rust- *Puccinia aristidae*



Puccinia

Spinach Rust- *Puccinia aristidae*

Rust is a heteroecious rust - Macrocyclic

Pycnial and aecial stages - Spinach, sugar beet

Uredial and telial - Various species of salt grass (*Distichlis* spp.) and needle grass (*Aristida* spp.)

Symptoms:

- **Also infects sugar beets**
- Light yellow - green spots - turns yellow pustules
- As rust cups open - turn orange

Management :

- Eradication of salt grass
- Burning crop residue
- Spray- Maneb, Zineb

Spinach Mosaic- Cucumber mosaic virus (CMV)

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Spinach Mosaic- Cucumber mosaic virus (CMV)

Symptoms:

- Younger leaves -slight chlorosis, narrow -"puckered" young leaves with inward rolling of margins
- In advanced stages –plants stunted
- Crown leaves may get blighted- death of growing point
- Symptoms develop faster at warm temperatures (>80°F)
- Disease severity varies with the virus strain, cultivar, crop age

Transmission: Vectors -aphids , cucumber beetles and mechanically by cultivation and handling

Management :

- Remove -wild cucumber vines, milkweed, and other weeds 1000 ft of the field
- Resistant cultivars are available from most seed companies.
- Do not grow spinach near cucurbit crops and tomatoes.
- Rogue the diseased plants seed crops to avoid infection seed